



CHAPTER

Instrumentation and Sensors

Pressure, flow, pH and conductivity measurement

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13.1 Overview and function

Inline sensors give the PLC (and the operator, via the HMI) a continuous read of the process: pressure, flow, pH and conductivity/level at key points in the pretreatment and treatment trains, plus a pressure gauge on each stack. All sensors use a standard 2-wire, loop-powered 4–20 mA output.



Figure 13.1. Suggested figure: photograph of the inline sensor array on a skid, with each sensor type (pressure, flow, pH, conductivity) labeled.

13.2 Key specifications

- Signal type: 2-wire, loop-powered, 4–20 mA.
- Confirmed sensor types in the pilot tag list: pressure (TSA_pressure), flow (TSA_Flow), pH (TSA_pH), conductivity/level (TSA_Cond, TSA_Level, TB_Cond, TB_Level, TE_Level).
- **Pressure:** WIKA A-10 pressure transmitter (gauge pressure, 4 mA = 0 bar; 2-wire, IP65/IP67, wetted parts 316L stainless steel, non-repeatability $\leq \pm 0.2\%$ of span). Stated maintenance-free by the manufacturer; clean only with water/mild detergent and a soft cloth — repairs are manufacturer-only.
- **pH/ORP:** candidate model GF DryLoc 2751 smart pH/ORP sensor electronics (2-wire, 4–20 mA).
- **Conductivity:** candidate GF 9900 transmitter with Type 2881–2884 conductivity electrodes.
- **Level:** 80 GHz radar level transmitters are under evaluation for tank level (e.g. Holykell HR2000, MDS-AM15) — TODO confirm whether radar or another level-sensing method is fitted on the standard product; these appear to be recent vendor quotes rather than a confirmed as-built spec.
- **Optional accessory:** HydroVolta's SonixED skid datasheet lists a **Cloud Multi-Parameter Water Analyzer (SUP-MPP1000)** as an available option — a 7-inch touchscreen instrument measuring turbidity, pH, conductivity, dissolved oxygen and residual chlorine, with RS485/4–20 mA output, for more detailed product-water quality monitoring than the standard sensor array.

TODO: confirm exact make/model per sensor type in the standard product.

13.3 Operation

Sensor readings feed the process-value displays described in chapter 6 and the alarm limits in appendix A.

13.4 Maintenance

No formal calibration-to-reference procedure (e.g. pH 4/7/10 buffer calibration) was found; what *is* documented is a real bench verification procedure used before commissioning any 2-wire 4–20 mA sensor:

1. Connect the sensor per the electrical schematic up to the terminal strip — **do not** connect to the PLC yet.
2. Insert a test resistor ($\sim 250\ \Omega$, giving a safe 1–5 V signal from the loop) and measure voltage with a voltage-only multimeter (e.g. Fluke 116).
3. For a pressure sensor (4 mA = 0 bar): $\sim 0.88\text{ V}$ across the test resistor confirms a normal 0 bar signal before connecting to the PLC.
4. For a pH sensor (0–14 pH, 4–20 mA) in a 7 pH reference solution: $\sim 2.64\text{ V}$ across a $220\ \Omega$ resistor (12 mA at mid-scale) is expected.
5. If the sensor checks out on the bench but still misbehaves once installed, the fault is in the field wiring, not the sensor.

TODO: this is a wiring/signal-integrity check, not a full calibration procedure — a genuine calibration interval/method for pH, conductivity and pressure sensors was not located.

13.5 Troubleshooting

Table 13.1. 4–20 mA sensor fault table

Symptom (bench test)	Likely cause	Remedy
0 V reading	No control voltage reaching sensor loop	Check DIN connector; confirm 24 V DC between terminal and PE; check polarity; confirm supply is on
Reading below 4 mA	Reversed polarity	Check polarity at the sensor connections
Reading stuck near 4 mA	Sensor connection contamination	Check the connection is dry and clean
Reading above 20 mA	Wiring fault / short	Check connections
No output signal at all (installed)	Cable break	Check loop continuity
Zero point has drifted	Overpressure limit exceeded, or wrong working temperature	Check against rated span and temperature range; recalibrate zero if within spec
Constant output despite pressure/process change	Mechanical fault in the sensor	Replace the sensor
Signal span varies over time	EMC interference	Shield the instrument/cable; remove the interference source
Signal span has dropped	Mechanical overload from an overpressure event	Replace the instrument
Fails on the bench too	Defective sensor	Replace the sensor
Passes bench test, fails installed	Field wiring fault	Inspect and repair wiring in the container

The first four rows above are from the real HydroVolta bench verification procedure; the following five are from the WIKA A-10 OEM manual's own fault-finding section and apply directly to pressure sensors, and by the same 2-wire 4–20 mA principle, are a reasonable first check for the pH and conductivity sensors as well.